Edith Gu

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EDUCATION

University of California San Diego — **San Diego, CA**
Master of Science, Data Science, GPA: 4.0 / 4.0 — Sep 2025 - Expected Jun 2027
Courses: Algorithm Analysis and Design, Data Structure, Data Base Principle, Machine Learning, Web Development, Distributed System, Database Management

Zhongnan University of Economics and Law — **Wuhan, China**
Bachelor of Science, Data Science, GPA: 3.9 / 4.0 — Sep 2021 - Jun 2025

TECHNICAL SKILLS

Programming Languages: Java, Python, Golang, JavaScript, SQL, C++
Web/Mobile: React, HTML/CSS, Express, Node.js, Android (Kotlin/Jetpack), Spring Boot, Microservices, Docker, Docker, AWS (RDS, ECR, App Runner), PostgreSQL, Room, Kafka, Kubernetes
Tools & Libraries: ElasticSearch, JWT, Git, Jetpack Compose, Hilt, Retrofit, OpenAI API, LangChain, Dall-E

PROFESSIONAL EXPERIENCE

Hubei Ronghai Investment Co., Ltd. — **Wuhan, China**
Software Engineer Intern — Jun. 2025 - Aug. 2025

- Built a high-concurrency mid-year promotion & order system for capital-market workflows, scaling throughput from 500 to 50K QPS (100×) via **Redis**, **RocketMQ** asynchronous pipelines, and **Spring Cloud** microservices.
- Refactored a legacy monolith into decoupled Promotion and Order services using **Spring Boot**, **Consul**, and **DDD**, improving deployment velocity, fault isolation, and service scalability under peak trading events.
- Implemented **TCC (Try-Confirm/Cancel)** distributed transactions via **Redis + Lua**, with **MySQL optimistic locking** and cache preheating, ensuring consistency and idempotency during high-load investment activities.
- Strengthened system stability and security with **Sentinel rate limiting**, **JMeter** stress testing, **CDN offloading**, and **anti-crawling/request protection**, maintaining SLA during promotional and financing surges.

Wuhan FraserGen Co., Ltd. — **Wuhan, China**
Software Engineer Intern — Feb. 2024 - Mar. 2024

- Engineered an internal **AI-powered bioinformatics assistant** using **LangChain, FastAPI, and WebSockets** to support natural language and voice-based interaction for genomics research workflows.
- Integrated **GPT-4-Turbo** for domain-aware reasoning and **Coqui TTS** for real-time speech synthesis, improving accessibility for cross-disciplinary teams within complex multi-modal genomics and clinical environments.
- Designed and implemented **long-term conversational memory** with **Redis**, enabling persistent research context across sessions, improving multi-turn task coherence by **~45%** in internal evaluations.
- Built a **RAG pipeline** using **OpenAI text-embedding-3-large** and **Qdrant**, reducing hallucinations by **~35%** and improving factual accuracy in scientific responses through optimized internal genomics document retrieval.
- Applied **recursive and semantic chunking** to optimize retrieval granularity for long, structured biological documents, minimizing embedding redundancy and improving retrieval precision for complex genomics queries.

PROJECTS EXPERIENCE

TitanKV — Distributed Key-Value Storage System

- Built a distributed key-value store in Java with **Multi-Raft consensus**, achieving 99.9% availability, **20K QPS**, and **<100ms P99** latency for linearizable reads/writes.
- Implemented **consistent hashing** with virtual nodes for 95% balanced data distribution and **dynamic shard rebalancing** with <1% performance overhead.
- Integrated **RocksDB (LSM-tree)** and **B+ Tree** hybrid engine with **MVCC** snapshot isolation; optimized reads via ReadIndex, FollowerRead, and async Apply batching, reducing latency by 35% and doubling throughput.

Foodly - Spring Boot & ReactJS Cloud-Based Food Ordering System

- Designed and implemented a full-stack food ordering system using **React** + **Ant Design** and **Spring Boot**, exposing optimized **RESTful APIs** with layered architecture ; reduced average API response latency by ~35% via DTO slimming, and **Caffeine**-backed hot-path caching.
- Optimized **PostgreSQL** data access and schema design on **AWS RDS**, applying indexed foreign keys, query plan analysis, and batched operations through **Spring Data JDBC**; improved high-traffic queries (cart retrieval, menu lookup) by 2–3×. Deployed containerized services with **Docker** + **AWS ECR** + App Runner.